DATASHEET

Circle Grid Calibration Target, Aluminum Composite

Circle grids use filled dot centroids rather than corner intersections. Centroid detection is more robust under defocus, low contrast and oblique viewing angles. Available in symmetric layout (grid of evenly spaced dots) and asymmetric layout (staggered rows for unambiguous orientation without additional markers). Both variants are supported by OpenCV (`cv::findCirclesGrid`). In HALCON, use them as a point-based calibration object; HALCON's own plate finder detects only HAL-* plates.

1 Ordering

Symmetric (SYM)

Part number	Size (W × H)	Circles (rows × cols)	Circle ⌀ / spacing	Total circles
CT-CIR-P-200X150-SYM	200 × 150 mm	5 × 8	12 / 20 mm	40
CT-CIR-P-300X200-SYM	300 × 200 mm	7 × 12	14 / 22 mm	84
CT-CIR-P-400X300-SYM	400 × 300 mm	10 × 14	17 / 25 mm	140
CT-CIR-P-600X400-SYM	600 × 400 mm	11 × 18	20 / 30 mm	198
CT-CIR-P-800X600-SYM	800 × 600 mm	16 × 22	22 / 33 mm	352

Asymmetric (ASYM)

Part number	Size (W × H)	Circles (rows × cols)	Circle ⌀ / spacing	Total circles
CT-CIR-P-200X150-ASY	200 × 150 mm	5 × 4	12 / 20 mm	20
CT-CIR-P-300X200-ASY	300 × 200 mm	7 × 6	14 / 22 mm	42
CT-CIR-P-400X300-ASY	400 × 300 mm	9 × 7	17 / 25 mm	63
CT-CIR-P-600X400-ASY	600 × 400 mm	11 × 9	20 / 30 mm	99
CT-CIR-P-800X600-ASY	800 × 600 mm	15 × 11	22 / 33 mm	165

Spacing is the pitch of the underlying square lattice. A symmetric grid has a circle at every lattice position; its `findCirclesGrid` pattern size is (cols, rows). An asymmetric grid has a circle at every other position, so circles in a row are two pitches apart and each row is shifted by one pitch against the next; rows × cols above are rows and circles per row, and the pattern size is (circles per row, rows) with `CALIB_CB_ASYMMETRIC_GRID`. Every layout leaves at least one pitch of white between the outermost circles and the board edge. Other grid sizes or circle dimensions: custom order (CUS).

Pattern area is centered with a white quiet zone ≥ 1 circle pitch on all sides; the edge strip with part number, serial and 100 mm scale bar sits outside the quiet zone.

2 Key specifications

ACCURACY	SUBSTRATE	FLATNESS
±0.3 mm + 0.2 mm/m (any circle center to nominal grid, at 20 °C)	Aluminum / LDPE / aluminum composite, 6 mm	≤ 0.5 mm per meter, board supported on three points

Unless stated otherwise, dimensions are nominal at 20 °C.

1 Specifications · geometry

FEATURE	SPECIFICATION
Feature position tolerance	±0.3 mm + 0.2 mm/m (any circle center to nominal grid, at 20 °C)
Circle pitch, mean	±0.05 mm
Circularity	≤ 0.05 mm radial deviation
Edge radius	R6 mm

2 Specifications · physical

FEATURE	SPECIFICATION
Substrate	Aluminum / LDPE / aluminum composite, 6 mm
Weight	6.6 kg/m² (300 × 200: ≈ 0.4 kg · 800 × 600: ≈ 3.2 kg)
Pattern	Direct UV print, matte, black on white; optical density ≥ 1.6
Flatness	≤ 0.5 mm per meter, board supported on three points
Mounting	4 × M4 threaded inserts on the rear (300 × 200 and larger); ¼″ fixture CT-ACC-M-FIX14 clamps any size

3 Specifications · environment

FEATURE	SPECIFICATION
Operating temperature	−50 to +80 °C
Thermal expansion	≈ 24 µm/(m·°C)
Storage	−20 to +60 °C, dry
Outdoor lifetime	up to 3 years (UV-stable print)
Cleaning	water or ≤ 70 % isopropyl alcohol, soft cloth

4 Deliverables

INCLUDED		SERVICE	
Measurement	Every board is measured before it ships (dimensional, flatness, visual)	Lead time	Made to order · ships in 2–3 weeks
Files	`.opencv.json`, `.kalibr.yaml` (symmetric only), `.ros2.yaml`, `.dxf`	Options	Logo print (-L) · hard case (CT-ACC-M-CSE400) · custom NRE
Documents	this datasheet, CT-UM-GEN user manual		

5 Compliance & customs

TOPIC	STATEMENT
RoHS	RoHS-compliant materials.
HS code	`9031 80` Not a toy. not for food contact.

6 Revision history

REV	DATE	CHANGE
A	2026-10	First issue, planning values

7 Disclaimer

Values are nominal unless stated as measured.